37) (I) AB $\rightarrow f(1,y) = 3\cos y$ em $-\pi/4 \leq y \leq \pi/4$;

$f'(1,y) = -3\,\text{sen}\, y = 0 \Rightarrow y = 0$;

$f(1,0) = 3$, $f(1,-\pi/4) = \frac{3\sqrt{2}}{2}$ e $f(1,\pi/4) = \frac{3\sqrt{2}}{2}$

(II) CD $\rightarrow f(3,y) = 3\cos y$ em $-\pi/4 \leq y \leq \pi/4$;

$f'(3,y) = -3\,\text{sen}\, y = 0 \Rightarrow y = 0$;

$f(3,0) = 3$, $f(3,-\pi/4) = \frac{3\sqrt{2}}{2}$ e $f(3,\pi/4) = \frac{3\sqrt{2}}{2}$

(III) BC $\rightarrow f(x,\pi/4) = \sqrt{2}/2\,(4x - x^2)$ em $1 \leq x \leq 3$;

$f'(x,\pi/4) = \sqrt{2}(2-x) = 0 \Rightarrow x = 2$;

$f(2,\pi/4) = 2$

(IV) AD $\rightarrow f(x,-\pi/4) = \frac{\sqrt{2}}{2}(4x - x^2)$ em $1 \leq x \leq 3$;

$f'(x,-\pi/4) = \sqrt{2}(2-x) = 0 \Rightarrow x = 2$

$f(2,-\pi/4) = 2\sqrt{2}$

(V) $\therefore$ o máx ABS é 4 em $(2,0)$ e o mín ABS é $2\sqrt{2}/2$ em $(3,-\pi/4)$, $(3,\pi/4)$, $(1,-\pi/4)$, $(1,\pi/4)$